

## 04\_stellarscope\_convergence

August 31, 2026

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[1]: import re
import matplotlib.pyplot as plt
import pandas as pd

[4]: log_path = "/mnt/transfer/TEbenchmarking/logs/stellarscope/"
# 1. Read the log file
# Replace 'stellarscope.log' with the actual path to your log file
dataset = "simulated_mm_RA"
sample = "young"
runid = "_260519_seed213"

with open(log_path + dataset+ "_" + sample + "_resume_pseudobulk_manualrun" +
    ↪runid + ".txt", 'r') as f:
    log_text = f.read()

# 2. Extract Iteration and Diff values using Regular Expressions
pattern = r"Iteration (\d+), ln1= ([\d.e+-]+), diff=([\d.e+-]+)"
matches = re.findall(pattern, log_text)

# 3. Convert to a Pandas DataFrame for easy handling
df = pd.DataFrame(matches, columns=["Iteration", "Ln1", "Diff"])
df["Iteration"] = df["Iteration"].astype(int)
df["Ln1"] = df["Ln1"].astype(float)
df["Diff"] = df["Diff"].astype(float)

# 4. Create the plot
plt.figure(figsize=(10, 6))
plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
    ↪linestyle='-', linewidth=2, label="Parameter Change ($\Delta\pi$)")

# Style the plot
plt.title("Stellarscope EM Convergence Curve (Parameter Stability) - Scenario 1_
    ↪young", fontsize=14, fontweight='bold')
plt.xlabel("Iteration", fontsize=12)
plt.ylabel("Absolute Difference (diff)", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
```

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plt.yscale('log') # EM differences drop exponentially; a log scale makes it
    ↪ easy to read
plt.legend()

# Show or save the plot
plt.tight_layout()
#plt.savefig("em_convergence_curve.png", dpi=300)
plt.show()

# 4. Create the plot
plt.figure(figsize=(10, 6))
plt.plot(df["Iteration"], df["Ln1"], marker='o', color='#1f77b4',
    ↪ linestyle='--', linewidth=2)

# Style the plot
plt.title("Stellarscope EM - ln1 in Scenario 1 young", fontsize=14,
    ↪ fontweight='bold')
plt.xlabel("Iteration", fontsize=12)
plt.ylabel("ln1", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.yscale('log') # EM differences drop exponentially; a log scale makes it
    ↪ easy to read
plt.legend()

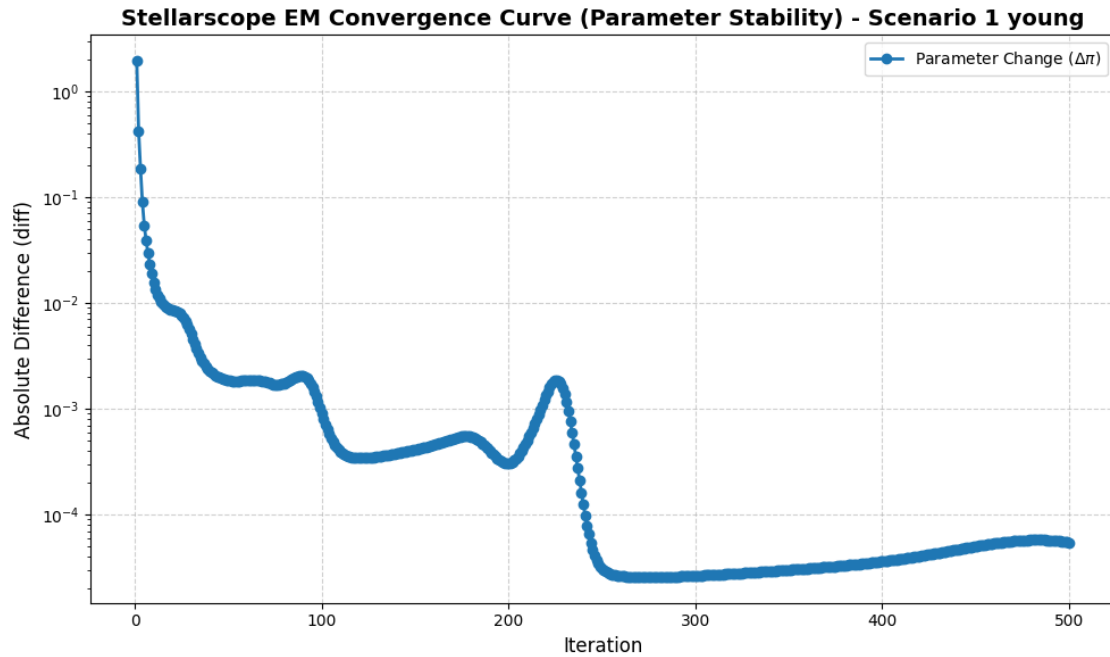
# Show or save the plot
plt.tight_layout()
#plt.savefig("em_convergence_curve.png", dpi=300)
plt.show()

```

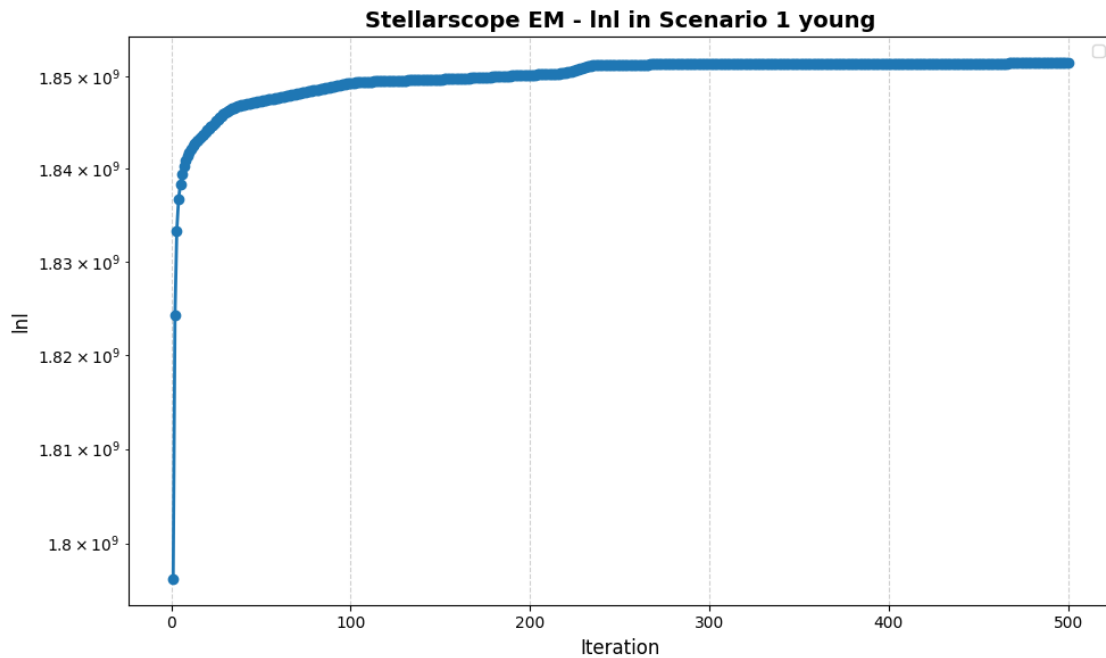
```

<>:23: SyntaxWarning: invalid escape sequence '\D'
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/mnt/transfer/tmp/ipykernel_1671607/2276970555.py:23: SyntaxWarning: invalid
escape sequence '\D'
    plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
linestyle='--', linewidth=2, label="Parameter Change ( $\Delta\pi$ )")

```



/mnt/transfer/tmp/ipykernel\_1671607/2276970555.py:49: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.  
 plt.legend()



```

[5]: log_path = "/mnt/transfer/TEbenchmarking/logs/stellarscope/"
# 1. Read the log file
# Replace 'stellarscope.log' with the actual path to your log file
dataset = "simulated_mm_RA"
sample = "young"
runid = "_260519"

with open(log_path + dataset + "_" + sample + "_resume_pseudobulk_manualrun" +
    ↪runid + ".txt", 'r') as f:
    log_text = f.read()

# 2. Extract Iteration and Diff values using Regular Expressions
pattern = r"Iteration (\d+), ln1=([\d.e+-]+), diff=([\d.e+-]+)"
matches = re.findall(pattern, log_text)

# 3. Convert to a Pandas DataFrame for easy handling
df = pd.DataFrame(matches, columns=["Iteration", "Ln1", "Diff"])
df["Iteration"] = df["Iteration"].astype(int)
df["Ln1"] = df["Ln1"].astype(float)
df["Diff"] = df["Diff"].astype(float)

# 4. Create the plot
plt.figure(figsize=(10, 6))
plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
    ↪linestyle='-', linewidth=2, label="Parameter Change ($\Delta\pi$)")

# Style the plot
plt.title("Stellarscope EM Convergence Curve (Parameter Stability) - Scenario 1_
    ↪young", fontsize=14, fontweight='bold')
plt.xlabel("Iteration", fontsize=12)
plt.ylabel("Absolute Difference (diff)", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.yscale('log') # EM differences drop exponentially; a log scale makes it
    ↪easy to read
plt.legend()

# Show or save the plot
plt.tight_layout()
#plt.savefig("em_convergence_curve.png", dpi=300)
plt.show()

# 4. Create the plot
plt.figure(figsize=(10, 6))
plt.plot(df["Iteration"], df["Ln1"], marker='o', color='#1f77b4',
    ↪linestyle='-', linewidth=2)

```

```

# Style the plot
plt.title("Stellarscope EM - ln1 in Scenario 1 young", fontsize=14,
         fontweight='bold')
plt.xlabel("Iteration", fontsize=12)
plt.ylabel("ln1", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.yscale('log') # EM differences drop exponentially; a log scale makes it
                 easy to read
plt.legend()

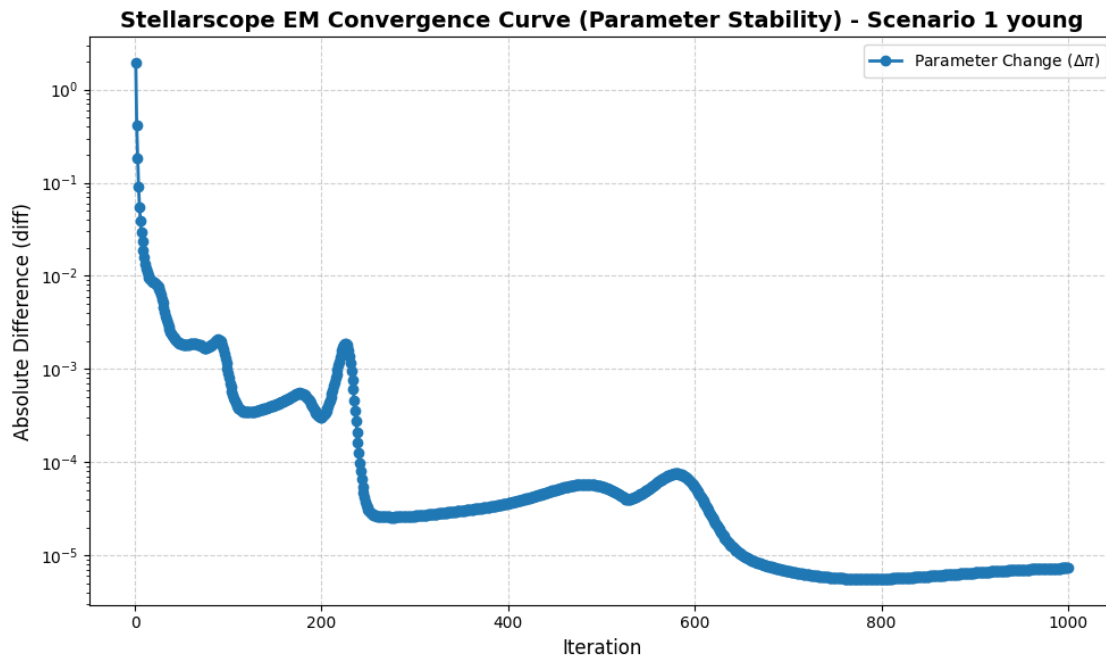
# Show or save the plot
plt.tight_layout()
#plt.savefig("em_convergence_curve.png", dpi=300)
plt.show()

```

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<>:23: SyntaxWarning: invalid escape sequence '\D'
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/mnt/transfer/tmp/ipykernel_1671607/1620144527.py:23: SyntaxWarning: invalid
escape sequence '\D'
    plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
linestyle='-', linewidth=2, label="Parameter Change ( $\Delta\pi$ )")

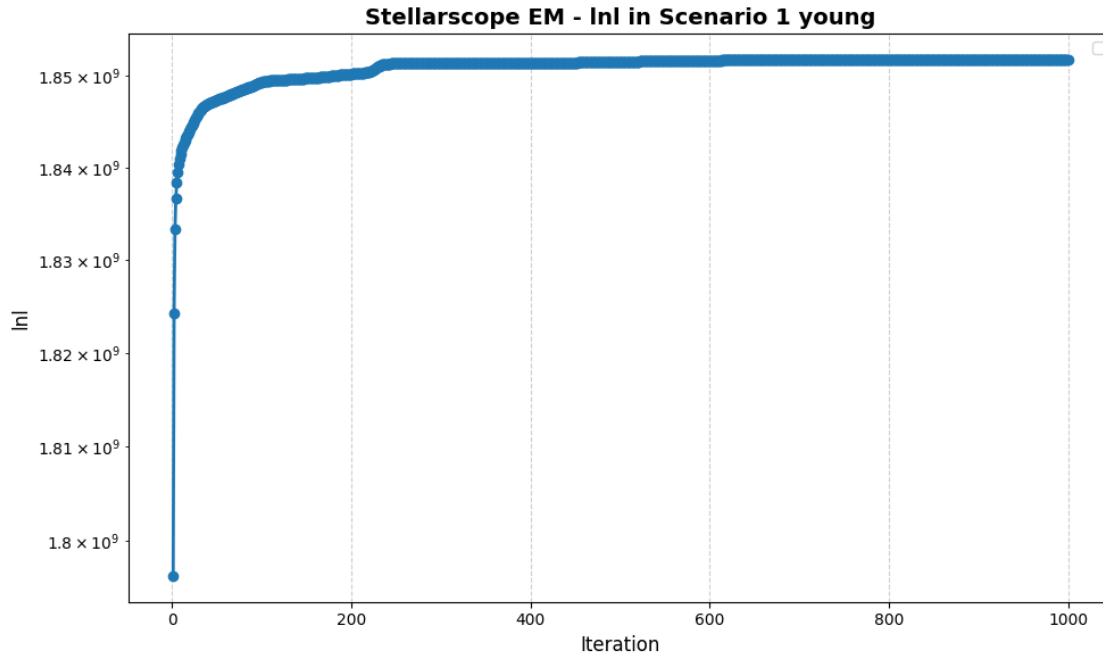
```



```

/mnt/transfer/tmp/ipykernel_1671607/1620144527.py:49: UserWarning: No artists
with labels found to put in legend. Note that artists whose label start with an
underscore are ignored when legend() is called with no argument.
    plt.legend()

```



```
[7]: log_path = "/mnt/transfer/TEbenchmarking/logs/stellarscope/"
# 1. Read the log file
# Replace 'stellarscope.log' with the actual path to your log file
dataset = "simulated_mm_RA"
sample = "young"
runid = "_seed1577742087_thr09"

with open(log_path + dataset+ "_" + sample + "_resume_pseudobulk_manualrun" +
runid + ".txt", 'r') as f:
    log_text = f.read()

# 2. Extract Iteration and Diff values using Regular Expressions
pattern = r"Iteration (\d+), diff=([\d.e+-]+)"
matches = re.findall(pattern, log_text)

# 3. Convert to a Pandas DataFrame for easy handling
df = pd.DataFrame(matches, columns=["Iteration", "Diff"])
df["Iteration"] = df["Iteration"].astype(int)
df["Diff"] = df["Diff"].astype(float)

# 4. Create the plot
plt.figure(figsize=(10, 6))
plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
linestyle='-', linewidth=2, label="Parameter Change ($\Delta\pi$)")
```

```

# Style the plot
plt.title("Stellarscope EM Convergence Curve - " + dataset, fontsize=14,
fontweight='bold')
plt.xlabel("Iteration", fontsize=12)
plt.ylabel("Absolute Difference (diff)", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.yscale('log') # EM differences drop exponentially; a log scale makes it
easy to read
plt.legend()

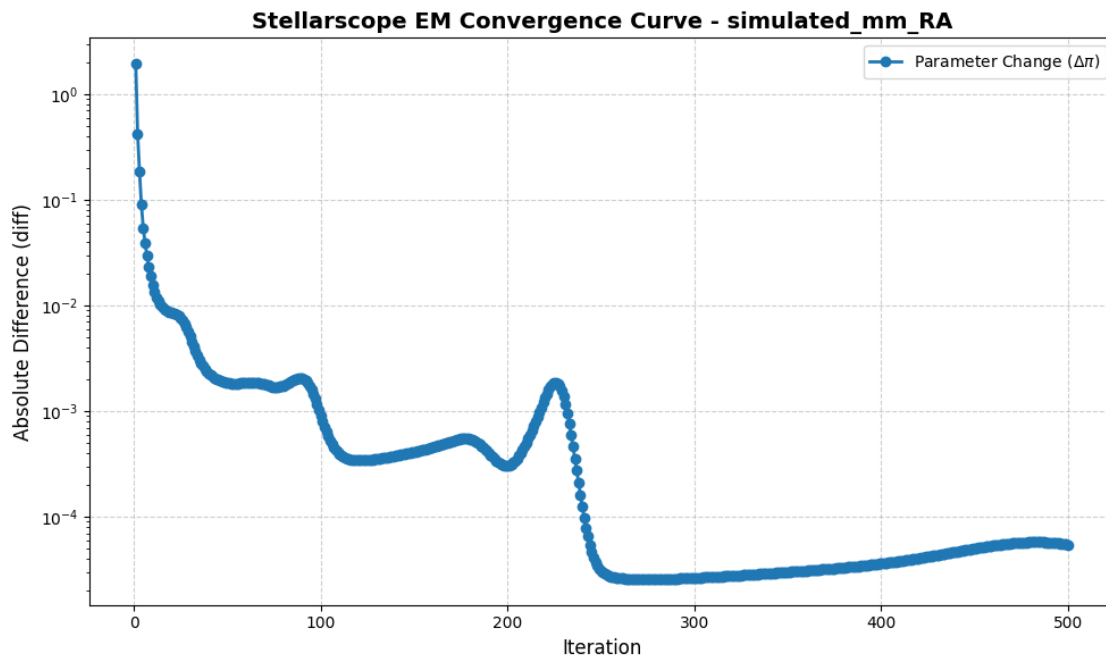
# Show or save the plot
plt.tight_layout()
#plt.savefig("em_convergence_curve.png", dpi=300)
plt.show()

```

```

<>:23: SyntaxWarning: invalid escape sequence '\D'
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/mnt/transfer/tmp/ipykernel_1671607/219263173.py:23: SyntaxWarning: invalid
escape sequence '\D'
    plt.plot(df["Iteration"], df["Diff"], marker='o', color='#1f77b4',
linestyle='-', linewidth=2, label="Parameter Change ( $\Delta\pi$ )")

```



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